Introduction to statistical inference

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What is statistical inference?

- Learning about what we do not observe (e.g. parameters) using what we observe (data)

  - **Is the coin fair? (p=0.5 or not)**
  - **How to collect data (x1,x2,…)**
  - **How to handle with the collected data? (sample statistic)**

Basic elements in statistical inference

- **Task** – parameter estimation, hypothesis test, and etc.
- **Experiment design**– data collection (e.g. simple random sampling, uniform experiment design, and etc.)
- **Data analysis**– sample statistic, pivot, and etc.
- **Evaluation**– consistency, unbiasedness, robustness, and etc.

Sample statistic

- A statistic or sample statistic is any quantity computed from values in a sample, e.g. sample mean, sample variance and etc. In other words, it is a function of sample.

Example: cognitive function and vitamin D

- Estimation: What is the average serum vitamin D in the whole population?
  - Sample statistic: mean vitamin D levels
- Hypothesis test: Are vitamin D levels and cognitive function correlated?
  - Sample statistic: correlation coefficient between vitamin D and cognitive function, measured by the Digit Symbol Substitution Test (DSST).

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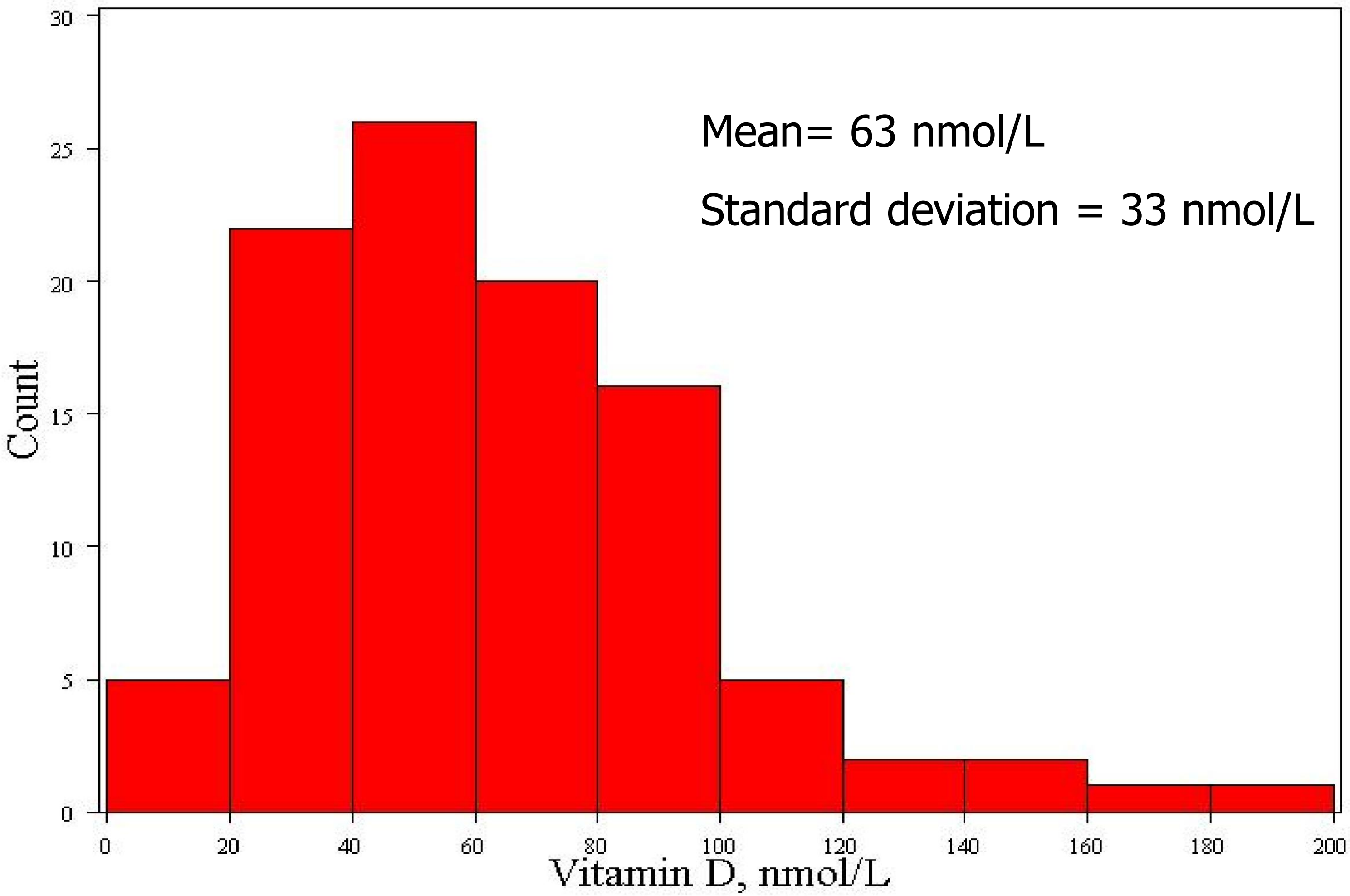
Sample distribution of vitamin D



Distribution of a sample statistic

- Statisticians ask a thought experiment: how much would the value of the statistic fluctuate if one could repeat a particular study over and over again with different samples of the same size?
- By answering this question, statisticians are able to pinpoint exactly how much uncertainty is associated with a given statistic.

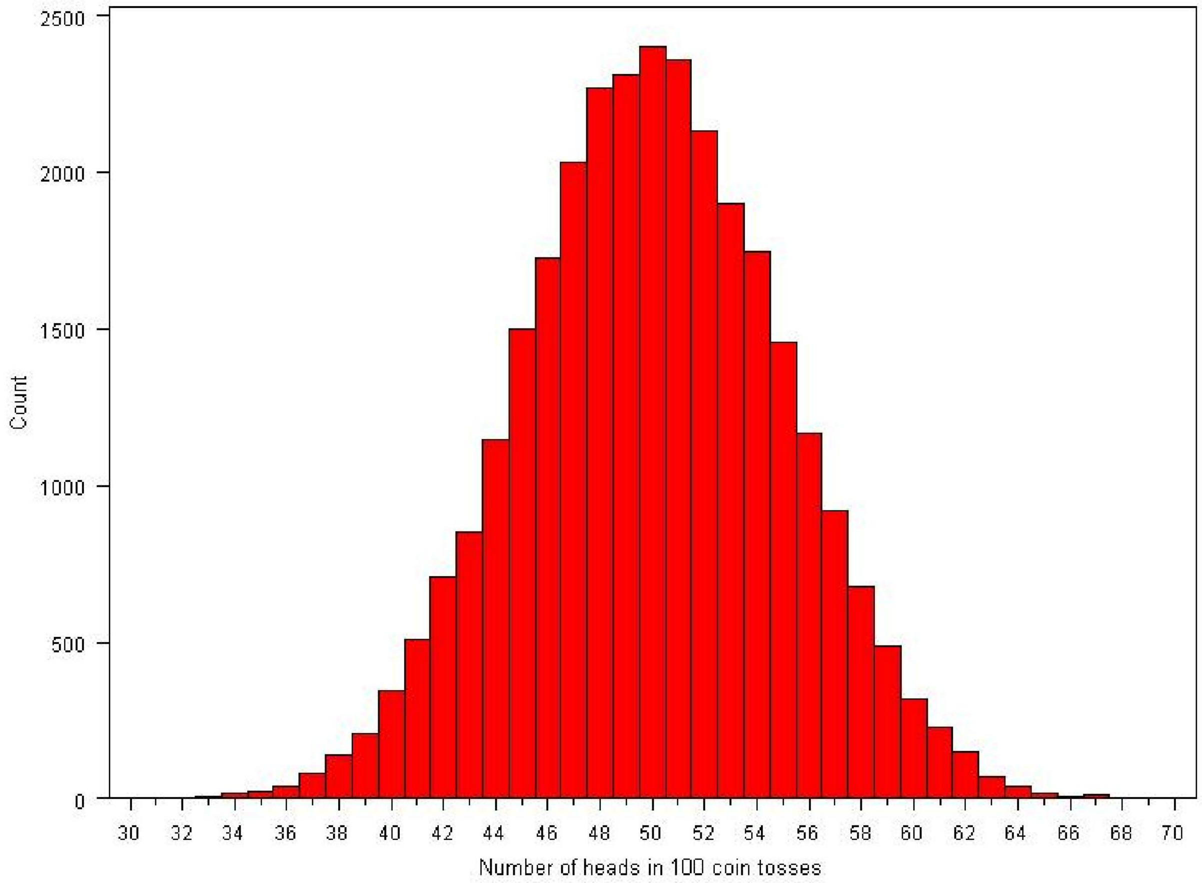
Distribution of a statistic

- Two approaches to determine the distribution of a statistic:
  - 1. (Computer) simulation
  - 2. Probability theory

Example of simulation

- How many heads come up in 100 coin tosses?
- Flip coins virtually
  - Flip a coin 100 times; count the number of heads.
  - Repeat this over and over again a large number of times (we'll try 30,000 repeats!)
  - Plot the 30,000 results.

Coin tosses…



Conclusions:

We usually get between 40 and 60 heads when we flip a coin 100 times.

It's extremely unlikely that we will get 30 heads or 70 heads.

Simple Random Samples

- A simple random sample (SRS) of size *n* consists of *n* individuals from the population chosen in such a way that every set of *n* individuals has an equal chance to be the sample actually selected, i.e.

$$(X_1, X_2, \ldots, X_n) \quad \text{(i.i.d)}$$

$$\Downarrow$$

$$(x_1, x_2, \ldots, x_n) \quad \text{(a realization)}$$

Probability theory

$$E(\bar{X}_n) = E\left(\frac{\sum_{i=1}^{n} X_i}{n}\right) = \frac{\sum_{i=1}^{n} E(X_i)}{n} = \frac{nE(X)}{n} = E(X)$$

$$Var(\bar{X}_n) = Var\left(\frac{\sum_{i=1}^{n} X_i}{n}\right) = \frac{\sum_{i=1}^{n} Var(X_i)}{n^2} = \frac{nVar(X)}{n^2} = \frac{Var(X)}{n}$$

The Central Limit Theorem

If all possible random samples, each of size *n*, are taken from any population with a mean μ and a standard deviation σ, the distribution of the sample means will:

1. have mean: $\boxed{\mu_{\bar{x}} = \mu}$

2. have standard deviation: $\boxed{\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}}$

3. be **approximately normally distributed** regardless of the shape of the parent population (normality improves with larger *n*)

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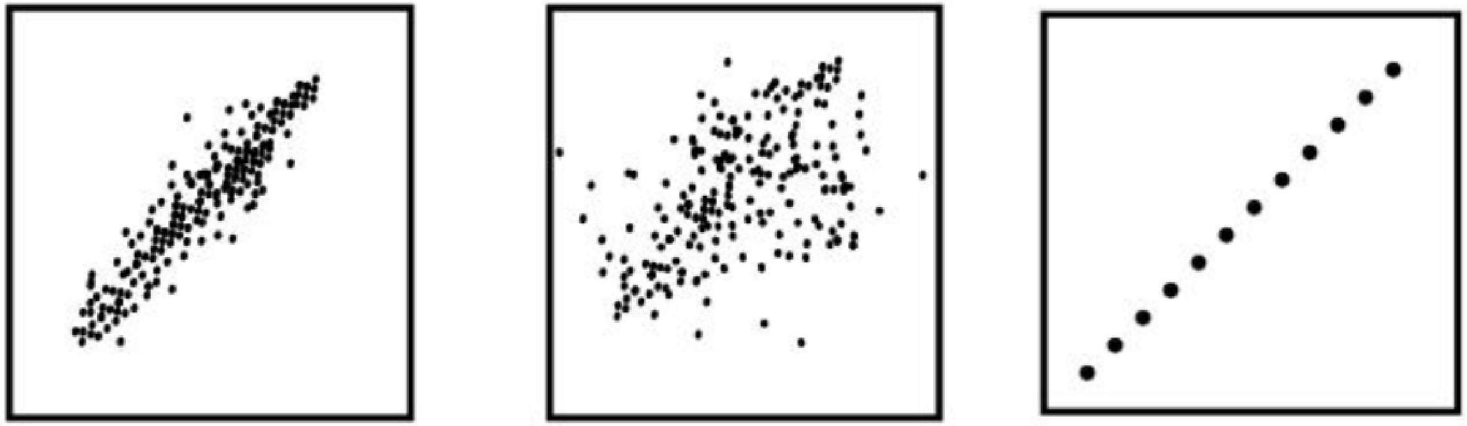
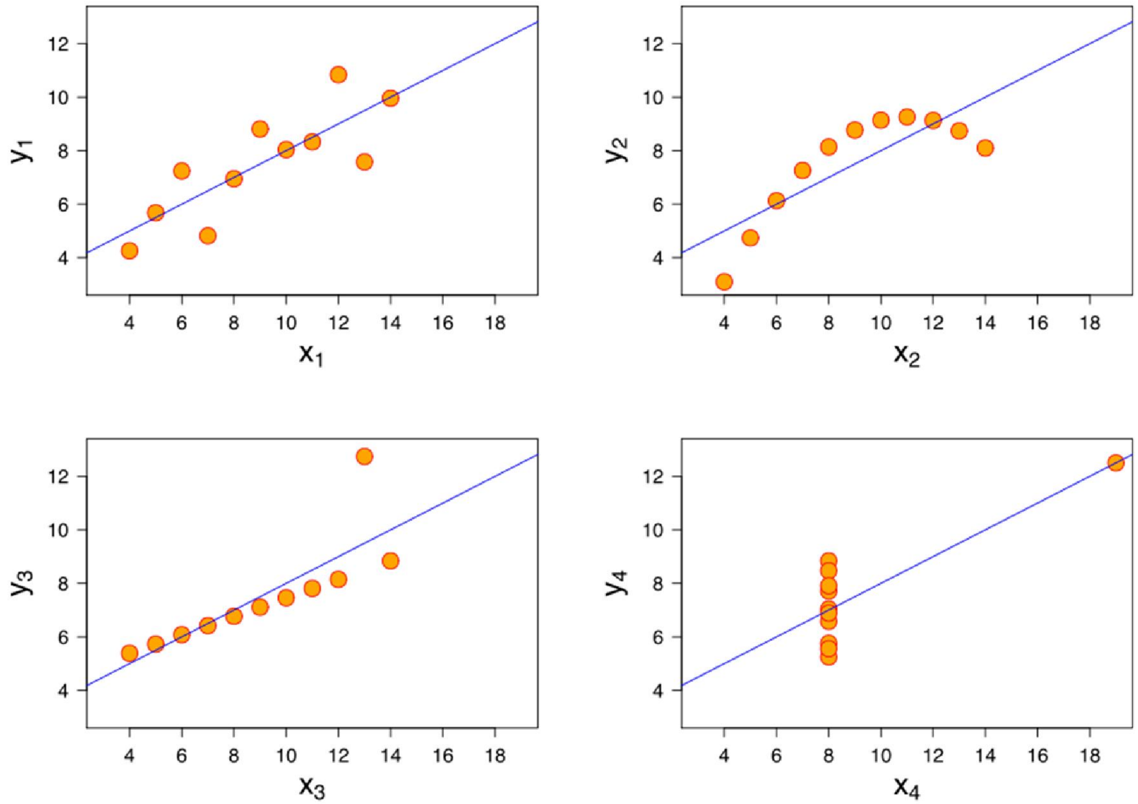
Pearson’s correlation coefficient

$$r = \frac{\sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{i=1}^{n} (X_i - \overline{X})^2} \sqrt{\sum_{i=1}^{n} (Y_i - \overline{Y})^2}}.$$



r = 0.85 r = 0.4 r = 1.0

Anscomb's Quartet



Distribution of a correlation coefficient

- 1. Shape of the distribution
  - Normally distributed for large samples
  - T-distribution for small samples (n<100)
- 2. Mean = true correlation coefficient

$$r = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

- 3. Standard error $\approx \dfrac{\sqrt{1-r^2}}{\sqrt{n}}$

Is cognitive function correlated with vitamin D?

Hypothesis test

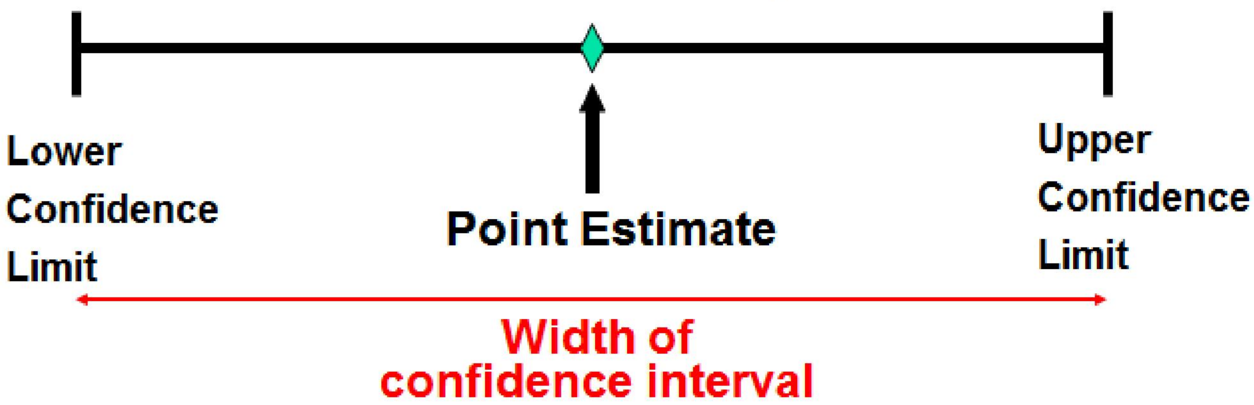
- Null hypothesis: $r = 0$
- Alternative hypothesis: $r \neq 0$

Two major tasks in statistical inferences

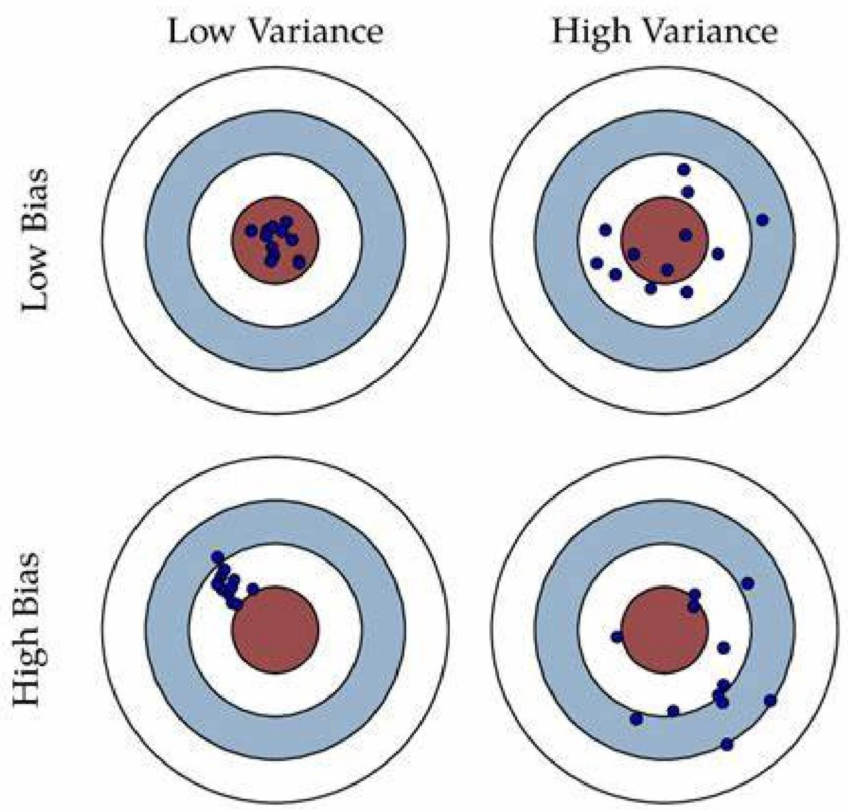
- Parameter estimation involves using sample data to estimate the parameters of a distribution.
- Hypothesis test is a way to test whether your results are valid by figuring out the odds that your results have happened by chance. If your results may have happened by chance, the experiment won't be repeatable and so has little use.

Parameter estimation

- Point estimation
- Interval estimation



What is a good estimator?



What is a good estimator?

$$X \sim N(\mu, \sigma^2)$$

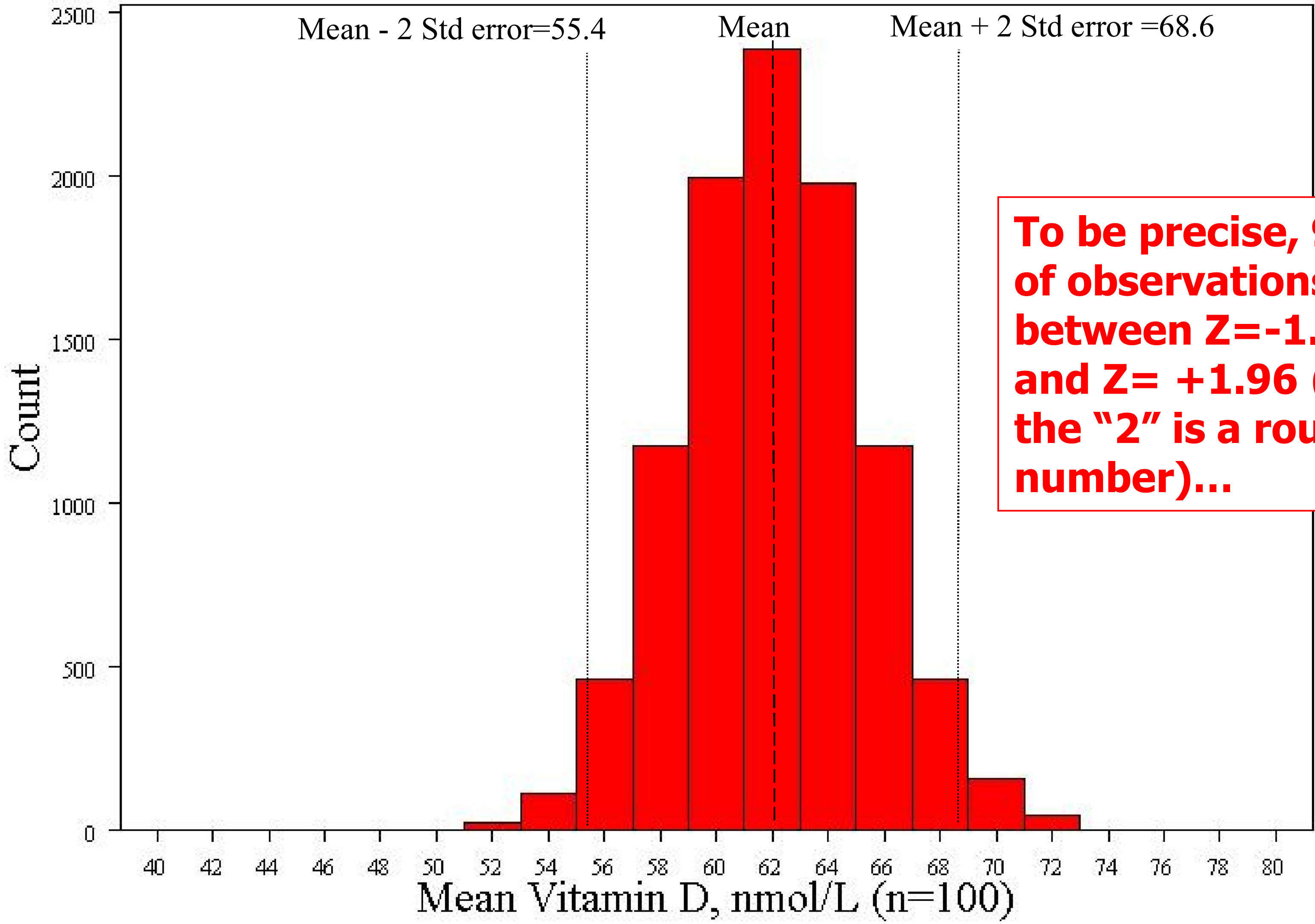
$$\boxed{E(\bar{X}_n) = \mu \quad \text{(unbiased)}}$$

$$\boxed{Var(\bar{X}_n) = \frac{\sigma^2}{n}}$$

Estimation (confidence intervals)

- Goal: capture the true value (e.g., the true mean) most of the time.
- A 95% confidence interval should include the true value about 95% of the time.
- A 99% confidence interval should include the true value about 99% of the time.

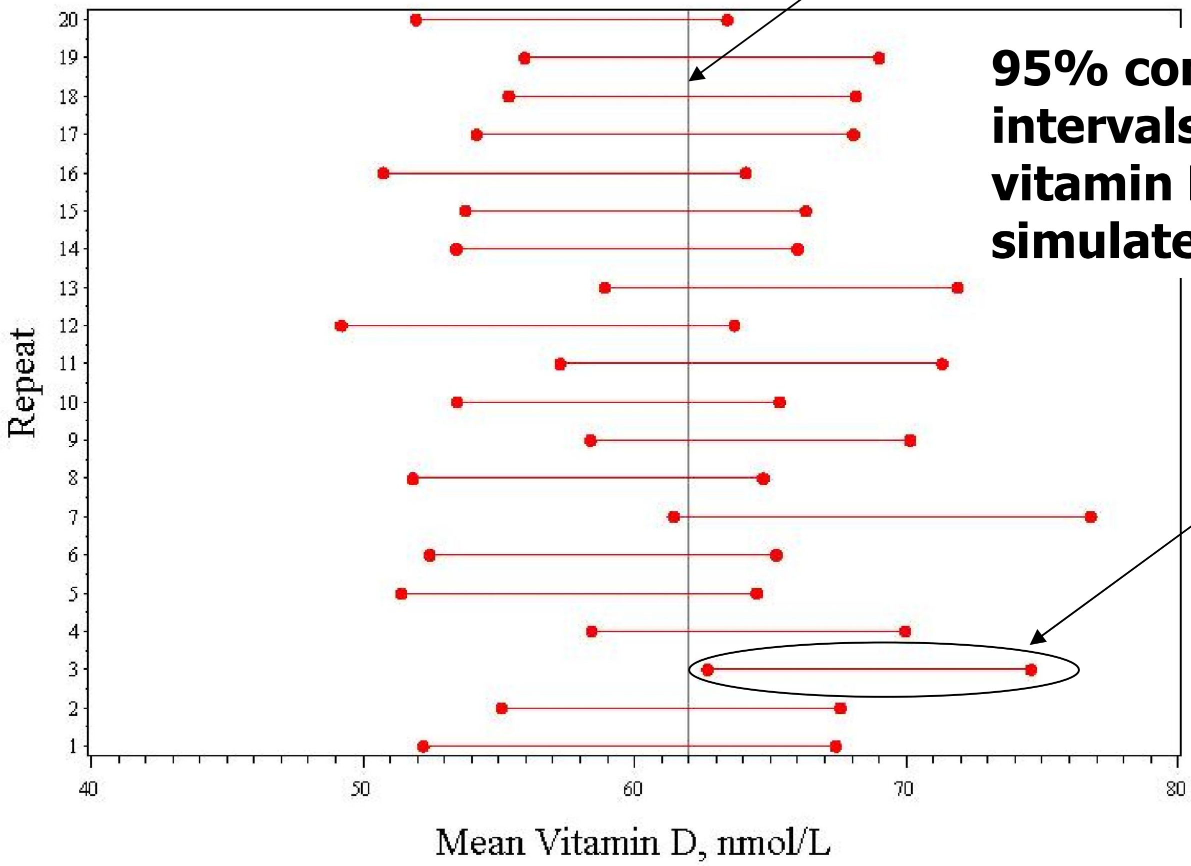
Recall: 68-95-99.7 rule for normal distributions! These is a 95% chance that the sample mean will fall within two standard errors of the true mean= 62 +/- 2*3.3 = 55.4 nmol/L to 68.6 nmol/L



To be precise, 95% of observations fall between Z=-1.96 and Z= +1.96 (so the "2" is a rounded number)...

Simulation of 20 studies of 100 men…

Vertical line indicates the true mean (62)



95% confidence intervals for the mean vitamin D for each of the simulated studies.

Only 1 confidence interval missed the true mean.

Confidence Intervals give:

- A plausible range of values for a population parameter.
- The precision of an estimate.(When sampling variability is high, the confidence interval will be wide to reflect the uncertainty of the observation.)
- Statistical significance (if the 95% CI does not cross the null value, it is significant at .05)

Testing Hypotheses

- 1. Is the mean vitamin D in middle-aged and older European men lower than 100 nmol/L (the "desirable" level)?
- 2. Is cognitive function correlated with vitamin D?

Is the mean vitamin D different than 100?

- Start by assuming that the mean = 100
- This is the "null hypothesis"
- Determine the distribution of statistics assuming that the null is true…

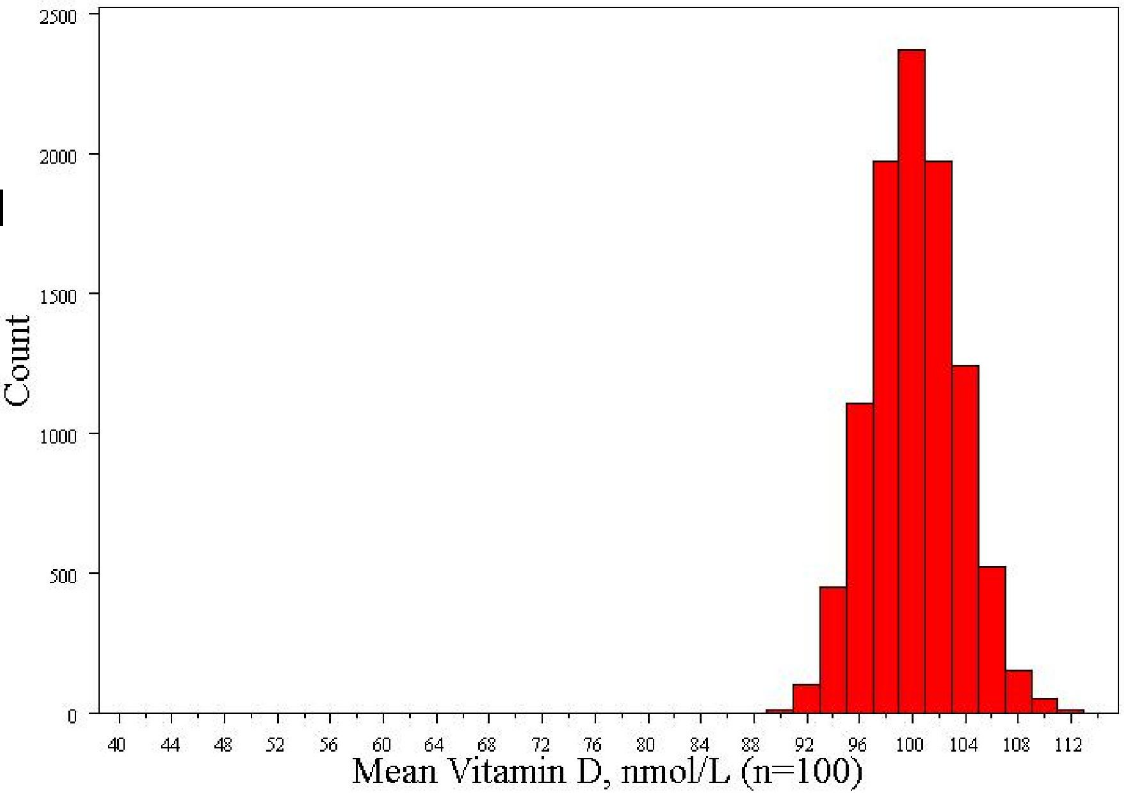
Computer simulation (10,000 repeats)…

This is called the null distribution!

Normally distributed

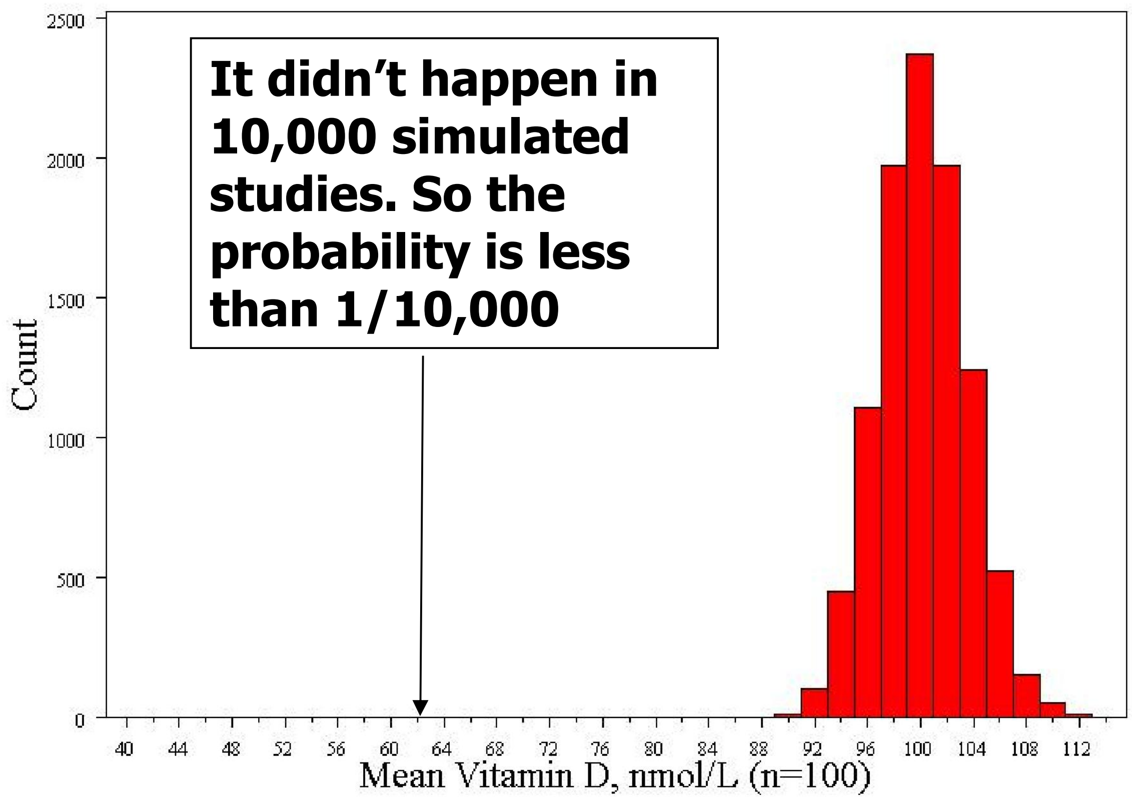
Std error = 3.3

Mean = 100



Compare the null distribution to the observed value…

What's the probability of seeing a sample mean of 63 nmol/L if the true mean is 100 nmol/L?



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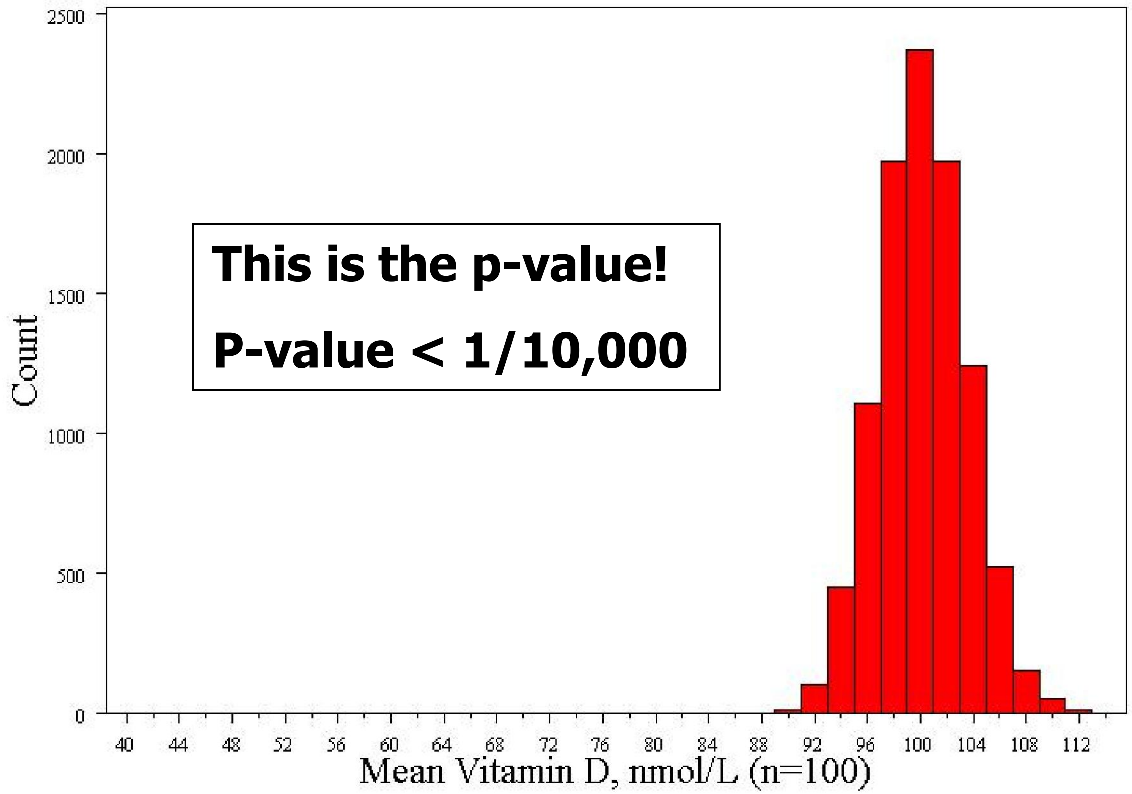


Calculating the p-value with a formula…

Because we know how normal curves work, we can exactly calculate the probability of seeing an average of 63 nmol/L if the true average weight is 100 (i.e., if our null hypothesis is true):

$$\boxed{Z = \frac{63-100}{3.3} = 11.2}$$

Z= 11.2, P-value << .0001

The P-value

P-value is the probability that we would have seen our data (or something more unexpected) just by chance if the null hypothesis (null value) is true.

Small p-values mean the null value is unlikely given our data.

Our data are so unlikely given the null hypothesis (<<1/10,000) that I'm going to reject the null hypothesis! (Don't want to reject our data!)

P-value<.0001 means:

The probability of seeing what you saw or something more extreme *if the null hypothesis is true (due to chance)*<.0001

P(empirical data/null hypothesis) <.0001

The P-value

- By convention, p-values of $<.05$ are often accepted as “statistically significant” in the literature; but this is an arbitrary cut-off.

- A cut-off of $p<.05$ means that in about 5 of 100 experiments, a result would appear significant just by chance (“Type I error”).

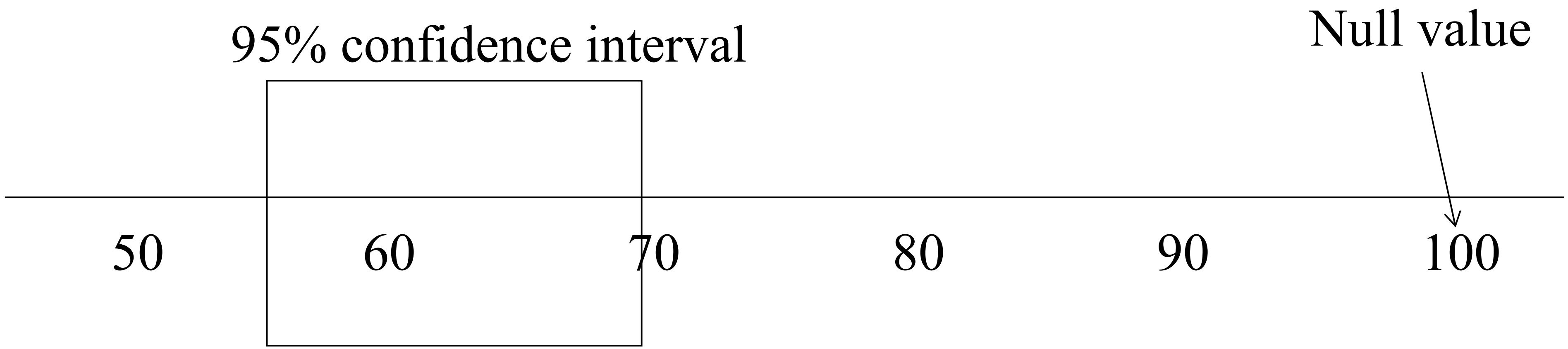
Summary: Hypothesis Test

The Steps:

1. Define your hypotheses (null, alternative)
2. Specify your null distribution
3. Do an experiment
4. Calculate the p-value of what you observed
5. Reject or fail to reject (~accept) the null hypothesis

- Confidence intervals give the same information (and more) than hypothesis tests…

Duality with hypothesis tests.



Null hypothesis: Average vitamin D is 100 nmol/L

Alternative hypothesis: Average vitamin D is not 100 nmol/L (two-sided)

P-value < .05

2. Is cognitive function correlated with vitamin D?

- Null hypothesis: $r = 0$
- Alternative hypothesis: $r \neq 0$
  - Two-sided hypothesis

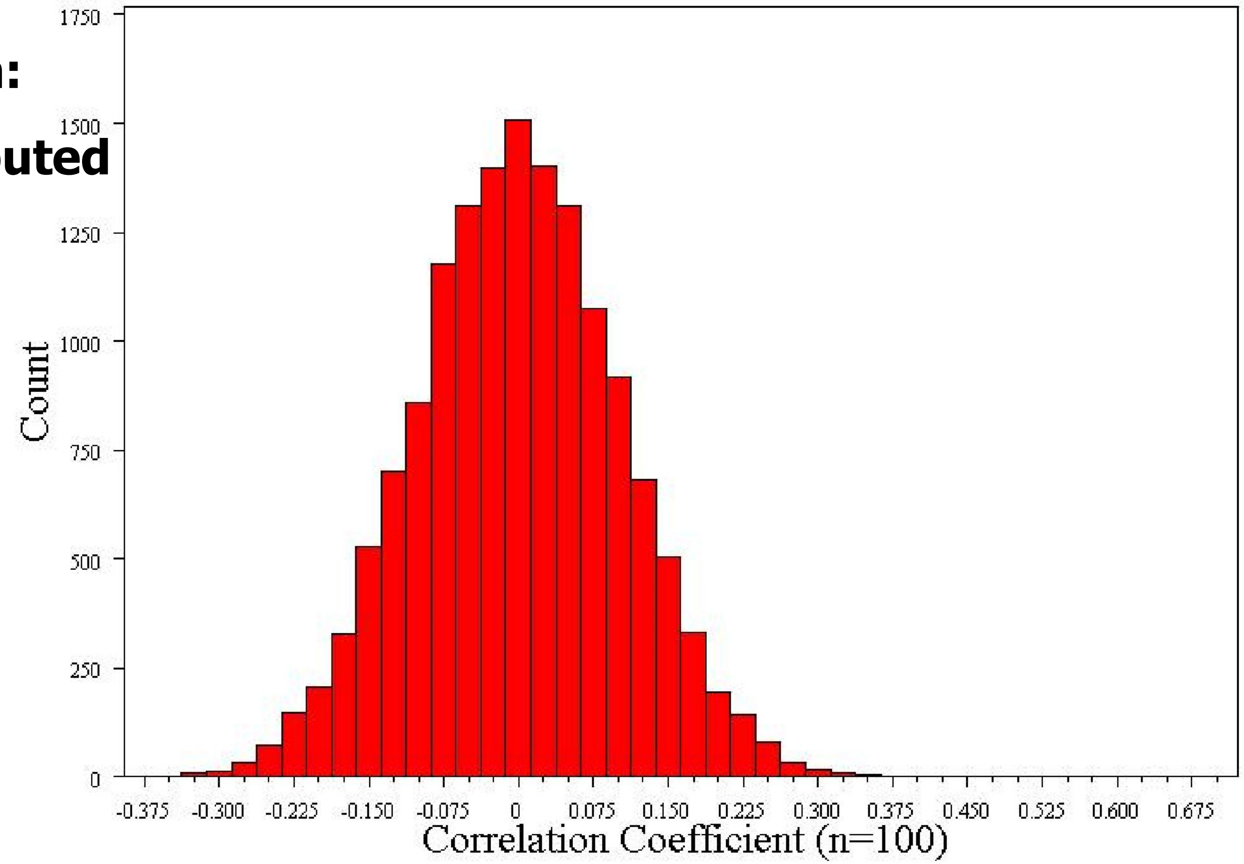
Computer simulation (15,000 repeats)…

Null distribution:

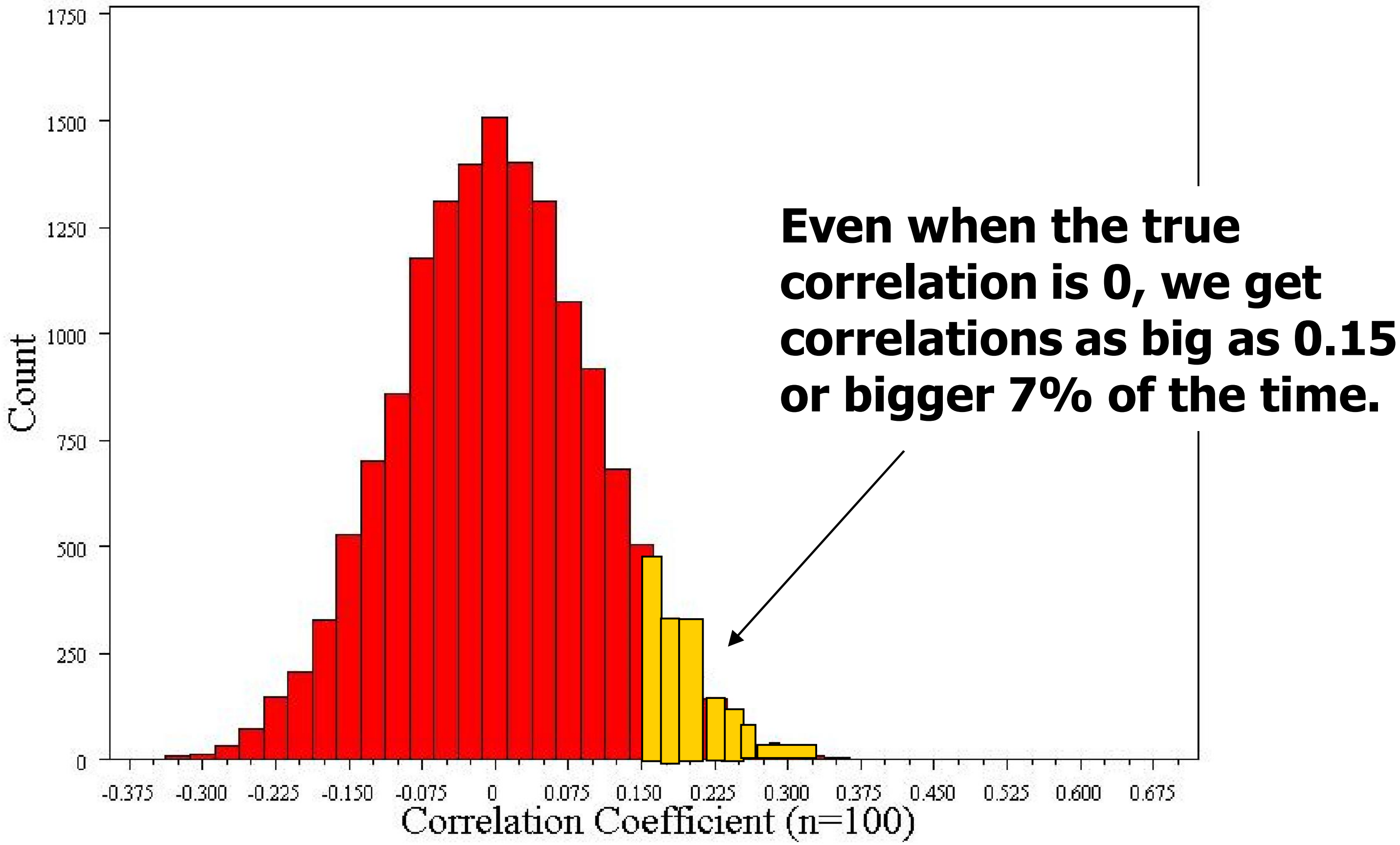
Normally distributed

Std error = 0.1

Mean = 0



What's the probability of our data?



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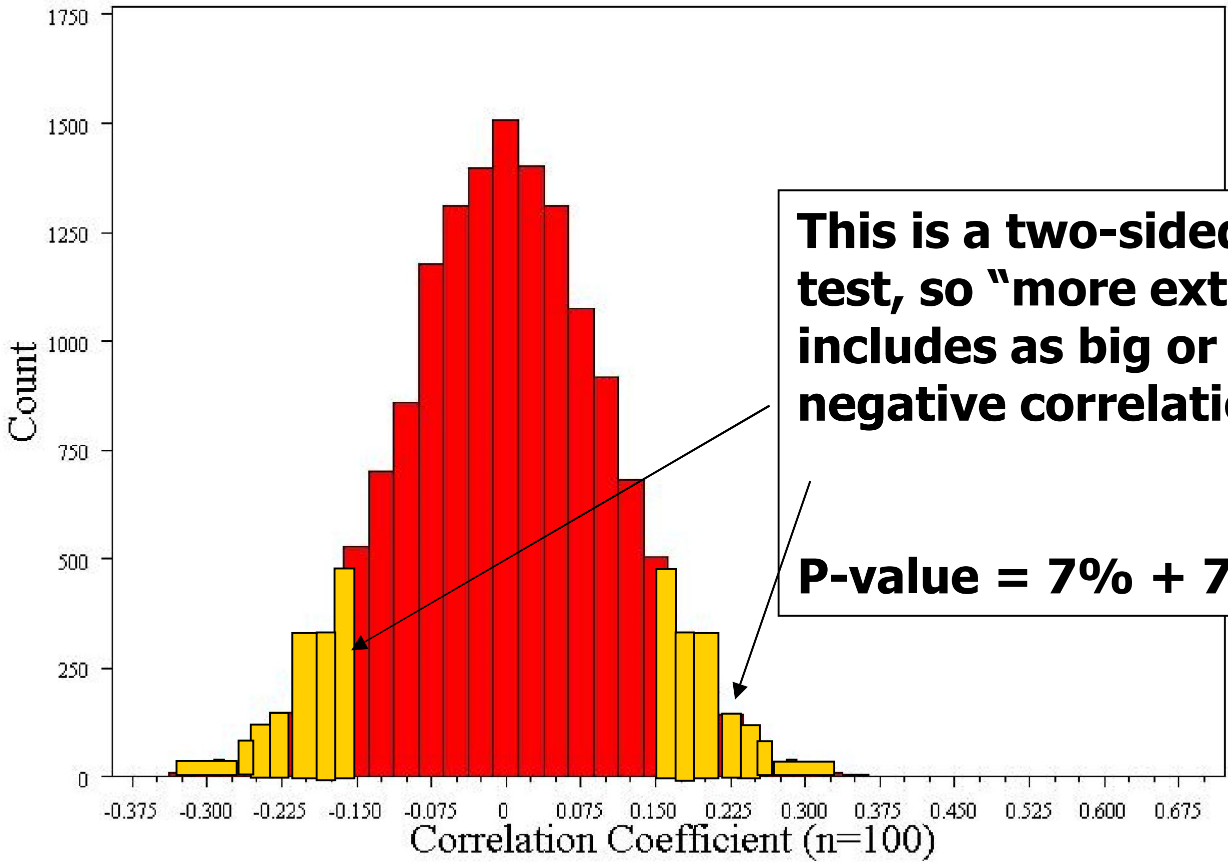
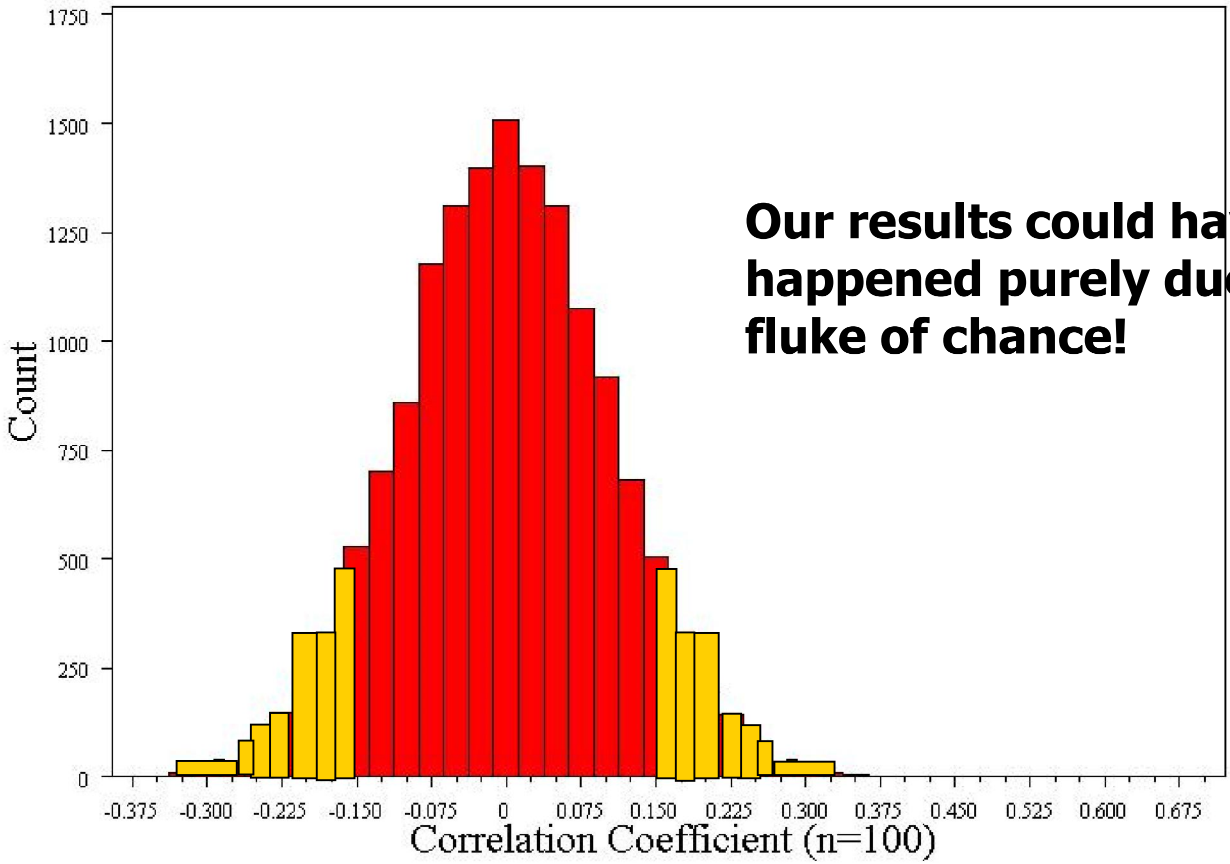


This is a two-sided hypothesis test, so "more extreme" includes as big or bigger negative correlations (<-0.15).

P-value = 7% + 7% = 14%

What's the probability of our data?



Our results could have happened purely due to a fluke of chance!

Formal hypothesis test

- 1. Null hypothesis: r=0
  - Alternative: $r \neq 0$ (two-sided)
- 2. Determine the null distribution
  - Normally distributed
  - Standard error = 0.1
- 3. Collect Data, r=0.15
- 4. Calculate the p-value for the data:
  - $$Z = \frac{0.15 - 0}{.1} = 1.5$$ **Z of 1.5 corresponds to a two-sided p-value of 14%**
- 5. Reject or fail to reject the null (fail to reject)

Or use confidence interval to gauge statistical significance…

- 95% CI = -0.05 to 0.35
- Thus, 0 (the null value) is a plausible value!
- P>.05